Empirically Calibrated Intervals for the n-th Prime: Construction, Measurement, and Limits

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Abstract

We describe in full a computational procedure that, given the primes up to N, produces an interval containing the n-th prime $p(n)$ for any $n \in (N, 2N]$. Across eight scales from $n = 10^8$ to $n = 6 \times 10^{15}$ the interval contained the true value in every case, and was 107 to 3,608 times narrower than the tightest proven bounds [1]; an exhaustive test at smaller scales covered 3,840,000 consecutive targets with a 100% containment rate.

This paper documents the construction completely: every equation, the measured basis for every constant, the alternatives that were tried and rejected with the numbers that rejected them, and the questions that remain unanswered. We treat the negative results as part of the contribution, since several of them close directions that would otherwise look promising.

The central limitation is stated at the outset and repeated where it matters: the interval is empirical, not proven. The calibration constant extrapolates a maximum-value statistic, and no theorem relates future maxima to past maxima. The only unconditional component is the Axler bound, applied as a clipping ceiling.

Keywords: n-th prime, prime counting function, Riemann R function, zeta zeros, empirical bounds, computational number theory
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1 Introduction

1.1 The two established approaches

Two distinct traditions address the location of $p(n)$.

Explicit bounds. Rosser and Schoenfeld [8] gave the first effective estimates. Dusart [4] and then Axler [1] tightened them successively. Axler's Theorem 1.1, valid for $n \geq 46{,}254{,}381$, states

$$p(n) < n\left(\ln n + \ln\ln n - 1 + \frac{\ln\ln n - 2}{\ln n} - \frac{\ln^2\ln n - 6\ln\ln n + 10.667}{2\ln^2 n}\right), \tag{1}$$

and Theorem 1.2, valid for $n \geq 2$, gives the matching lower bound with the constant 10.667 replaced by 11.508 and the inequality reversed.

These are unconditional. Their weakness is width: at $n = 2 \times 10^{15}$ the interval spans about 1.3×10^{12}.

Exponentially tight effective bounds. A recent development in the same direction is due to Visser [9], who derives bounds of the form

$$\mathrm{li}^{-1}\Big(n\left[1 - a(\ln[n\ln n])^{b+1}e^{-c\sqrt{\ln[n\ln n]}}\right]\Big) < p(n) < \mathrm{li}^{-1}\Big(n\left[1 + a(\ln[n\ln n])^{b+1}e^{-c\sqrt{\ln[n\ln n]}}\right]\Big) \tag{2}$$

from effective bounds on $|\pi(x) - \mathrm{li}(x)|$. These are structurally close to the present work: both localise $p(n)$ by inverting a counting function rather than by expanding an asymptotic series. We compare against them in §10.2.

Exact computation. The Meissel–Lehmer family [7, 3, 5], implemented in primecount [10], computes $\pi(x)$ and hence $p(n)$ exactly. On ordinary hardware $p(10^{12})$ returns in milliseconds.

1.2 What this work is, and what it is not

This is not a competitor to exact computation. Where primecount applies it is faster and exact, and two of the four operating modes described below call it internally.

Nor is it a theorem. The interval produced carries no proof.

What it is: a direct measurement of how far the best available analytic estimates deviate from the truth, and an interval calibrated to that measurement. An exact algorithm returns a number with no uncertainty description. A proven bound describes uncertainty conservatively. This work measures the deviation and reports the interval that follows.

1.3 Operating range

Training on $[1, N]$ yields a valid containment on $(N, 2N]$ and no further. Every result reported here is at a ratio of exactly 2.00.

This is not an incidental restriction. The calibration constant C_{EMP} (§5) is measured on the training range and applied beyond it; the further one extrapolates, the weaker the justification. The single-doubling choice keeps the extrapolation at a distance where it has been tested.

1.4 How to read this paper

Sections 3–7 build the construction, each component with its derivation and the measurement that justifies its form. Section 9 describes the four modes. Section 10 reports results. Sections 11–15 document what failed and what remains unresolved; these are not appendices to the contribution but part of it.

Numbers reported as "measured" are from runs whose parameters are given at the point of use. Where a measurement was made on an earlier or modified version of the code, this is stated.

2 Notation and conventions

Symbol	Meaning
$p(n)$	the n-th prime
$\pi(x)$	the number of primes $\leq x$
$\mathrm{li}(x)$	the logarithmic integral, $\int_0^x dt/\ln t$ (Cauchy p.v.)
$R(x)$	the Riemann prime counting function, (3)
$\mu(k)$	the Möbius function
γ_k	the imaginary part of the k-th non-trivial zeta zero
N	the training bound; the primes in $[1, N]$ are known
$F_b(n)$	the analytic base estimator, (12)
$F_z(n)$	the full estimator including the zeta layer, (13)
$r(n)$	the residual $p(n) - F_z(n)$
C_{EMP}	the empirical deviation constant, (14)
$[L_n, U_n]$	the Axler interval at n
$[x_1, x_2]$	the interval produced by the procedure

Throughout, "the model" refers to the complete procedure, and "residual" to $r(n)$ unless qualified.

3 The analytic base

3.1 Why the Riemann function rather than the logarithmic integral

The prime number theorem gives $\pi(x) \sim \mathrm{li}(x)$, and one might invert li to estimate $p(n)$. This is not the best available choice, and the reason is structural rather than empirical.

Riemann's function

$$R(x) = \sum_{k \geq 1} \frac{\mu(k)}{k} \, \mathrm{li}\Big(x^{1/k}\Big) \tag{3}$$

arises from Möbius inversion of

$$J(x) = \sum_{k \geq 1} \frac{1}{k} \pi\Big(x^{1/k}\Big) = \pi(x) + \tfrac{1}{2}\pi\Big(x^{1/2}\Big) + \tfrac{1}{3}\pi\Big(x^{1/3}\Big) + \cdots, \tag{4}$$

the function for which li is the natural approximation. In other words, $\mathrm{li}(x)$ approximates a count that includes prime powers; $R(x)$ removes that excess and approximates $\pi(x)$ directly.

The excess is not negligible at the scales of interest. At $x = 10^{12}$, the term $\frac{1}{2}\pi(x^{1/2}) \approx 39{,}250$, which is comparable to the entire deviation we are trying to bound.

Measured effect. Over 400 targets outside the training range, using each estimator as the base with no further correction:

Base	Mean |error|	Ratio
$\mathrm{li}^{-1}(n)$	47,075	—
$R^{-1}(n)$	6,381	7.4×

A further measurement across scales shows the advantage is not constant: R^{-1} was between 1.4× and 62× closer than li^{-1} depending on the scale, the advantage being largest where the prime-power correction is proportionally largest.

3.2 Computing R^{-1} at scale

Direct evaluation of (3) requires roughly seven evaluations of Ei (terms with $\mu(k) = 0$ vanish; the series is truncated once $x^{1/k} < 2$). Over the hundreds of millions of points required for a full C_{EMP} scan (§5) this is prohibitive.

We exploit the smoothness of the difference. Define

$$D(x) = R(x) - \mathrm{li}(x) = \sum_{k \geq 2} \frac{\mu(k)}{k} \, \mathrm{li}\Big(x^{1/k}\Big). \tag{5}$$

This is dominated by $-\frac{1}{2}\,\mathrm{li}(x^{1/2})$, which is smooth and slowly varying. We tabulate D once on a fixed logarithmic grid of 200,000 points spanning $[2, 10^{18}]$ and interpolate linearly.

Observation 1. The interpolation error of D on this grid is at most about 8×10^{-6}, against residuals of order 10^3–10^4. It is therefore not a limiting factor.

Remark 1 (Why the grid range must be fixed). An earlier implementation sized the grid to the current problem. This made results depend on the run: the same n evaluated in two different sessions differed by about 10^{-6}, which propagated into the branching logic (§7) and produced different intervals. The grid range is now a compile-time constant.

Inversion is by Newton's method on $R(x) = n$, using $R'(x) = 1/\ln x$:

$$x_{j+1} = x_j - (R(x_j) - n)\ln x_j. \tag{6}$$

The first step is free. Seeding with $x_0 = \mathrm{li}^{-1}(n)$ gives $\mathrm{li}(x_0) = n$ exactly, hence $R(x_0) - n = D(x_0)$ by (5), so

$$x_1 = x_0 - D(x_0)\ln x_0 \tag{7}$$

requires no evaluation of Ei at all. A single further step with the true li suffices:

Algorithm 1 $R^{-1}(n)$

1: $x \leftarrow \mathrm{li}^{-1}(n)$ ▷ seed; Newton on li
2: $x \leftarrow x - D(x)\ln x$ ▷ free step, uses tabulated D
3: $x \leftarrow x - (\mathrm{li}(x) + D(x) - n)\ln x$ ▷ one Ei evaluation
4: return x

3.3 A cheaper inverse for the calibration scan

The C_{EMP} scan (§5) evaluates R^{-1} at every integer up to N. Even one Ei call per point is costly there.

For that loop only, we replace li in the early Newton iterations by its asymptotic series

$$\mathrm{li}(x) \approx \frac{x}{\ln x}\left(1 + \frac{1}{\ln x} + \frac{2}{\ln^2 x} + \frac{6}{\ln^3 x} + \cdots\right), \tag{8}$$

which requires no special function, and apply a single correction with the true li at the end to remove the series bias.

Remark 2 (The series fails at small argument). Series (8) diverges from li for small x: measured error reaches 89% at $n = 10$. Points below a threshold are therefore computed by the exact route. Since they are few, the cost is negligible. With this guard, the maximum error of the fast inverse across a mixed range is about 10^{-12}, and it runs 2.2× faster than the exact version on large inputs.

4 The zeta-zero layer

4.1 Motivation from the explicit formula

Riemann's explicit formula expresses $J(x)$ in terms of the non-trivial zeros $\rho = \frac{1}{2} + i\gamma$:

$$J(x) = \mathrm{li}(x) - \sum_{\rho} \mathrm{li}(x^{\rho}) - \ln 2 + \int_x^{\infty} \frac{dt}{t(t^2-1)\ln t}. \tag{9}$$

Pairing conjugate zeros, each contributes an oscillation of the form

$$\mathrm{li}(x^{\rho}) + \mathrm{li}\big(x^{\bar{\rho}}\big) \;\approx\; \frac{2\sqrt{x}}{\ln x} \cdot \frac{\cos(\gamma \ln x - \arg)}{|\rho|}, \tag{10}$$

with amplitude decaying as $1/|\rho| \approx 1/\gamma$.

This motivates — but does not derive — our basis. We are estimating $p(n)$, not $\pi(x)$, and we apply the basis at $R^{-1}(n)$ rather than at x. The construction below should be read as a regression basis inspired by (10), not as a truncation of (9).

4.2 Construction

For each of the first $K = 60$ zeros we form the pair

$$A_k(x) = \frac{\sqrt{x}}{\gamma_k \ln x}, \tag{11}$$
$$\varphi_{2k}(x) = A_k(x)\cos(\gamma_k \ln x), \qquad \varphi_{2k+1}(x) = A_k(x)\sin(\gamma_k \ln x),$$

giving $2K = 120$ features. Writing $b = R^{-1}(n)$, the base estimator is

$$F_b(n) = b = R^{-1}(n), \tag{12}$$

and the full estimator is

$$F_z(n) = F_b(n) + \sum_{j=0}^{2K-1} \beta_j \varphi_j(b) + c_0. \tag{13}$$

The coefficients β are fitted by ridge regression on the residuals $p(n) - F_b(n)$ over a grid of the training range. The penalty is chosen by cross-validation over a logarithmic sweep.

4.3 The role of c_0

c_0 is the median of $p(n) - F_b(n) - \sum \beta_j \varphi_j$ over all grid points, computed after the regression.

Its purpose is to centre the estimator on the known sample. It matters that it is a median rather than a mean (robustness to the extreme residuals documented in §13), and that it is computed over all points rather than a subset — a distinction that becomes important in §11.9.

4.4 Measured effect

Over 400 targets outside the training range:

Estimator	Mean |error|	Ratio to li^{-1}
$\mathrm{li}^{-1}(n)$	47,075	—
$F_b(n) = R^{-1}(n)$	6,381	7.4×
$F_z(n)$, (13)	3,574	13.2×

The zeta layer contributes a factor of 1.79 over R^{-1} alone, and improves the estimate on 68.8% of the 400 targets.

4.5 Negative result: additional zeros degrade the estimate

The natural question is whether more zeros help. We computed zeros 61–1060 (up to $\gamma = 1489$) and repeated the fit, evaluating out of sample with a temporal split so that no target informed its own fit.

Zeros used	$\max \gamma$	Out-of-sample improvement
60 (baseline)	169	—
200	486	+4.99%
500	888	+2.90%
1,000	1,489	+1.19%

The improvement decreases monotonically as zeros are added. Two mechanisms combine:

1. By (11) the amplitude of the k-th feature scales as $1/\gamma_k$. The 1000th zero contributes at roughly 11% of the amplitude of the first, so its signal-to-noise ratio in a finite sample is correspondingly worse.

2. Ridge regression penalises all coefficients jointly. Adding 1880 low-amplitude features forces the cross-validated penalty upward, which suppresses the coefficients of the informative low zeros along with the uninformative high ones.

We therefore retain $K = 60$. We note this is a statement about this regression setup at these sample sizes, not about the explicit formula: in (9) the high zeros do carry information, and a different estimator might extract it.

5 The empirical bound

5.1 Definition

Definition 1. For a training bound N and a lower cutoff n_0,

$$C_{\mathrm{EMP}} = \max_{n_0 \le n \le N} \frac{|p(n) - F_z(n)|}{\ln^2 n}. \tag{14}$$

Two choices in (14) require justification: the normalisation $\ln^2 n$ and the cutoff n_0. Both are addressed below, and neither has a theoretical basis — a fact we consider important enough to state before giving the measurements that motivated them.

5.2 The normalisation has no theoretical justification

The factor $\ln^2 n$ is borrowed from Cramér's conjecture [2], which concerns the largest gap between consecutive primes near x:

$$\limsup_{n\to\infty} \frac{p_{n+1} - p_n}{\ln^2 p_n} = 1 \quad \text{(conjectural)}. \tag{15}$$

Two objections apply, and we state both:

1. Cramér's statement is a conjecture, not a theorem. Granville [6] has argued the constant is likely larger than 1.

2. More seriously, it describes prime gaps, whereas (14) measures model residuals. These are different quantities. A large residual can arise without a large gap, and conversely.

The normalisation is therefore an empirical choice that happens to have a suggestive form. Section 11.1 reports what happened when we tested alternatives — including one that is better motivated theoretically and performed worse.

5.3 The lower cutoff is critical

Measured at $N = 2 \times 10^6$:

Cutoff n_0	C_{EMP}	Location of the maximum
10	64.250	$n = 11$
100	16.800	$n = 111$
1,000	9.9	$n \approx 1{,}200$
10,000	7.553	$n = 1{,}887{,}313$
100,000	7.553	$n = 1{,}887{,}313$

At $n = 11$ we have $\ln^2 11 = 5.75$. The residual there is small in absolute terms; the ratio is large only because the denominator is trivially small. This is an artefact of the normalisation at small argument, not an extremal event.

From $n_0 = 10^4$ upward the value is stable, and the maximum falls at $n = 1{,}887{,}313$ — a location that corresponds to a genuine large prime gap. We adopt $n_0 = 10{,}000$.

Remark 3 (Impact of the error). An earlier version used $n_0 = 10$. This inflated C_{EMP} by a factor of 8.5, and hence the interval by the same factor, at every scale. The error was invisible in the containment rate — a too-wide interval still contains — and was found only by printing the location of the maximum rather than its value.

5.4 Structure of the maximum

C_{EMP} is a maximum-value statistic. Its behaviour as N grows determines whether extrapolating it is defensible, so we measured that behaviour directly.

Tracking the running maximum over 2×10^7 points:

Quantity	Value
Total updates to the running maximum	451
Number of clusters (updates within 10,000 of each other)	20
Median inter-cluster distance	207,454
Maximum inter-cluster distance	4,689,667
Coefficient of variation of inter-cluster distances	1.46

Observation 2. A coefficient of variation of 1.46 exceeds the value 1.0 expected of a Poisson process. The updates are therefore more clustered than random: long quiet stretches, of millions of points with no change at all, punctuated by bursts.

Two trends accompany increasing scale:

Correlation	r
Position vs. distance to the next cluster	$+0.52$
Position vs. size of the increment	-0.45

Clusters become rarer and increments smaller as N grows. This is encouraging for extrapolation, but Observation 2 prevents it from being conclusive: a clustered process can be quiet for a long time and then move sharply.

Remark 4 (Consequence for the margin). Observation 2 is, we believe, the reason no regular law for the safety margin could be derived (§12). One cannot fit a trend to a process whose increments arrive in rare bursts of variable size.

6 The two models and containment by intersection

6.1 Why two models

The estimator F_z of (13) gives a point. To obtain an interval one needs a second source of information, and the natural one is the inverse relation: while the $p(n)$ model asks "where is the n-th prime?", a $\pi(x)$ model asks "how many primes are below x?".

We build both. Each is a stochastic walker (§7) that maintains a set of branches; at any query point each yields not a single value but a set of admissible values — what we call the cloud.

6.2 The intersection

Write $\mathrm{cloud}_\pi(x) \subset \mathbb{N}$ for the set of counts the π model admits at x. The interval is

$$\begin{aligned} x_1 &= \inf\{x : n \in \mathrm{cloud}_\pi(x)\}, \\ x_2 &= \sup\{x : n \in \mathrm{cloud}_\pi(x)\}, \end{aligned} \tag{16}$$

subsequently clipped by the Axler bound:

$$[\max(L_n, x_1),\ \min(U_n, x_2)]. \tag{17}$$

The clipping in (17) is what makes the reported interval never worse than the proven one. Where the calibrated interval is narrower — which is the case at every scale reported — the result is empirical; where it would be wider, the proven bound takes over.

6.3 Why not a fixed envelope

The obvious alternative is to centre an envelope on the point estimate:

$$\left[F_z(n) - m\,C_{\mathrm{EMP}} \ln^2 n,\ F_z(n) + m\,C_{\mathrm{EMP}} \ln^2 n\right] \tag{18}$$

for some multiplier m. This is simpler, faster, and uses only one model. We implemented it and measured it against the intersection at scales where the two disagree:

	Intersection (16)	Envelope (18)
$N = 7 \times 10^6$	0 failures, min. margin 19.1%	1 failure, min. margin 0.53%
$N = 1.2 \times 10^7$	0 failures, min. margin 32.6%	4 failures, min. margin 0.19%

The envelope fails. The reason is that (18) is symmetric and uniform: it applies the same half-width regardless of local structure. The intersection is neither. It draws on a second, independent model, and the width it yields at a given n reflects how quickly cloud_π moves through n — which varies with the local density of primes.

Remark 5 (This decided a design question). Before this measurement it was not clear that the second model earned its cost; it roughly doubles the computation. The measurement settled it. We record it because the envelope is the natural first thing to try, and it looks adequate until tested at a scale where it is not.

6.4 Seed independence

The search in (16) is implemented by expanding outward from a starting point. Because cloud_π has gaps — regions where the admissible set excludes n — a naive search can stop at a gap and return a boundary that depends on where it started.

Measured: of four targets tested with different starting points, three gave different bounds, differing by 1–3%.

The fix is to run the search from several starting points distributed across the admissible region and take the union of the resulting intervals. We use 5 seeds, placed at evenly spaced quantiles of the admissible set; with this the result is stable across repeated runs. This is a correctness requirement, not an optimisation: without it the reported interval is not a well-defined function of n.

7 The branching mechanism

7.1 Structure

Both models advance in steps. The $p(n)$ model steps in n with spacing $\mathrm{CP}_{pn} = 40$; the π model steps in x with spacing $\mathrm{CP}_{\pi} \ln x$ where $\mathrm{CP}_{\pi} = 5$.

Each branch carries an offset o, a slope correction s, and two counters. At each checkpoint the observed local slope

$$\widehat{f}' = \frac{dF_b}{dn} + s + \frac{dW}{dn} \tag{19}$$

is compared against a reference slope t, and the branch takes one of four actions:

Condition	Action
$\left\lvert \widehat{f}' - t \right\rvert / \lvert t \rvert \le \theta$	hold: keep o
value above the upper threshold	brake: decrease o
value below the lower threshold	jump: increase o
otherwise	split into three branches

Splitting is capped: when the branch count exceeds MX the set is pruned to SEL by taking an evenly spaced subset in offset order, which preserves the extremes (which determine the interval) while bounding cost.

7.2 Thresholds

The upper and lower thresholds that separate brake and jump from splitting are set at $\pm T$ around the base, with

$$T = 2.28 \cdot N^{0.486} \cdot 0.84. \tag{20}$$

The exponent 0.486 was obtained by fitting the observed scale of residuals against N over the training range; the leading constant and the factor 0.84 were set during development and not subsequently optimised. We report (20) as it stands rather than as a derived quantity: it is a fitted form with two hand-set constants, and we have no argument that it is the right one.

The tolerance θ in the hold test starts at 0.02 and adapts with branch history. Writing c for the number of consecutive brakes or jumps a branch has taken and h for its consecutive holds:

$$\begin{aligned} \theta &\leftarrow \min\left(0.05,\ 0.02 \cdot 1.5^{\min(c,5)}\right) && \text{if } c \ge 3, \\ \theta &\leftarrow \max\left(0.005,\ \theta \cdot 0.85^{h}\right) && \text{if } h \ge 5, \end{aligned} \tag{21}$$

and the correction factor applied when the branch does act starts at 0.60 and decays as $\max(0.30,\ 0.60 \cdot 0.9^{c})$.

The intent is that a branch which has been correcting repeatedly is given room to continue, while one that has been stable is held to a tighter tolerance. The constants in (21) are design choices; we did not test alternatives.

7.3 The reference slope

The reference t for the $p(n)$ model derives from the lower convex hull of the points $(n, p(n))$ over the training range. Specifically, the hull is computed by the monotone chain algorithm, its edge slopes are extracted, and t is taken at the first index where consecutive integer parts of the slope differ by at least 2.

For the π model, $t = 1/\ln x$ directly, this being the derivative of li.

Remark 6 (Memory in the hull computation). The hull is over N points. A direct implementation materialising n and $p(n)$ as float arrays and pre-allocating the hull required about 20 GB at $N = 5 \times 10^8$ — an actual crash. The current implementation generates n internally, passes the primes in their stored integer form, and grows the hull dynamically. The result is bit-identical and about 67× faster.

7.4 The memory term

Each branch carries an exponentially decaying memory of the accumulated ridge correction:

$$w_{j+1} = \lambda\, w_j + (1 - \lambda)\, \hat{w}, \tag{22}$$

where $\hat{w}$ is a fresh estimate at the current point and λ is the measured autocorrelation of the residuals at the checkpoint spacing.

Remark 7 (A term that cancelled itself). An earlier form updated w toward the branch-mean of the current offsets. Since offsets are recentred to mean zero after each pruning, this drove w to zero: measured, it decayed from -46.6 to zero within ten checkpoints, so the memory term had no effect at all. The current form (22) targets a fresh independent estimate instead.

7.5 Measuring λ

λ in (22) is the lag-CP autocorrelation of the residual series. Measuring it correctly turned out to require care.

The natural implementation — take a sample of the checkpoints, then correlate adjacent sampled values — is wrong. After subsampling m points from a range of M checkpoints, adjacent samples are M/m checkpoints apart, not one. At $N = 5 \times 10^7$ with $m = 3000$ this measures a lag of about 16,680 instead of 40.

We verified the diagnosis on a synthetic AR(1) series with known correlation 0.85:

Method	Measured λ	Error
Subsample then correlate	-0.0355	0.886
Blocks of consecutive points	$+0.849$	0.001

The correct method takes short blocks of consecutive checkpoints, distributed across the range. This preserves the true lag while still sampling the whole range.

The same correction applies to the π side, where it also removed a substantial cost: the previous implementation walked every checkpoint from the first prime to N.

Method ($N = 5 \times 10^6$)	λ_π	Time
Full walk (1,000,138 points)	0.99846695	3.67 s
Blocks (12,000 points)	0.99849368	0.04 s

Agreement to 2.7×10^{-5}, 84× faster.

8 The anomaly mechanism

8.1 Description

Each branch runs a four-phase cycle intended to widen the cloud when the model appears to be under stress:

Phase	Trigger / duration	Action
0	$\lvert w\rvert/(C_{\mathrm{EMP}} \ln^2 n) \geq 0.30$	enter phase 1
1	2 checkpoints	add candidates at $\pm\lvert w\rvert$
2	1 checkpoint	transit
3	2 checkpoints	add candidates at $\mp 0.4\lvert w\rvert$, then return to 0

8.2 Measured effect: negligible on the output

We measured its effect by disabling it (setting the trigger threshold to 10^9) and re-running:

	Width	Contained	Min. margin
Enabled	6,911	8/8	26.7%
Disabled	6,901	8/8	26.7%

A difference of 0.14% in width and none at all in containment.

8.3 But real on the internal structure

The mechanism is not inert. Measured on the cloud itself rather than the interval:

Quantity	Change when enabled
Median of the candidate cloud	0.9% more accurate
Dispersion of the cloud	2.2% lower
Candidates surviving the π filter	19% more

Observation 3. The internal improvement does not reach the interval because the interval is determined by the two extreme candidates alone. Any change to the density of the cloud between those extremes is invisible in the output.

Observation 3 is a general property of the extraction rule, and it recurs in §11.2 and §11.4: several mechanisms that measurably improve the cloud leave the interval unchanged, for this reason.

8.4 Why it was retained

Removing it yields a 0.14% narrower interval — within measurement noise, and not worth a change to a component whose interaction with the rest of the branching is not fully characterised. It is retained, with a switch to disable it for measurement.

9 Operating modes

The four modes differ only in how C_{EMP} is obtained. The containment range, the estimator, the branching, and the intersection are identical in all four.

Mode	C_{EMP} source	Practical ceiling	Nature
A	Full sieve, stored	$\sim 2 \times 10^9$ (memory)	Measured
B	Sieve in chunks, discarded	$\sim 2 \times 10^{11}$ (time)	Measured
C	10^5-point sample via primecount	$\sim 2 \times 10^{13}$	Measured + margin
F	Extrapolation from prior scales	Tool limit	Estimated

9.1 Mode A: full sieve

The primes are generated by a segmented sieve and held in memory. Every $n \in [n_0, N]$ is inspected, so C_{EMP} is exact for the training range.

Storage is the constraint. Stored as 64-bit integers, 10^9 primes occupy 8 GB.

Compressed storage. We store gaps rather than values. Every prime after 2 is odd, so gaps between consecutive primes are even, and gap/2 fits in one byte: the largest known gap below 2.3×10^{10} is about 464, giving gap/2 = 232 < 255. With anchor points at fixed stride, any slice is reconstructed by a cumulative sum from the nearest anchor.

Memory at $N = 10^9$: 8 GB → 1 GB. Reconstructed slices are bit-identical to the original array.

Remark 8 (An edge case worth recording). The gap $2 \to 3$ equals 1, which is odd — 2 being the only even prime. Halving it corrupts the representation. The prime 2 is therefore stored separately and compression begins at 3.

Remark 9 (Peak memory, not final memory). An initial implementation generated all primes and then compressed. This leaves the peak at $8N$ bytes, which is where the crash occurs; compression after the peak does not help. Compression now happens per sieve segment, so the peak is the segment size.

9.2 Mode B: streaming

The sieve runs in chunks; C_{EMP} is updated on each chunk as it is produced, and the chunk is discarded. Every point is still inspected, so coverage of the training range is complete — but memory is the chunk size, about 8 MB, independent of N.

Observation 4. Mode B was verified to produce C_{EMP} identical to the last bit against mode A. It is therefore mode A without the memory ceiling, and is the preferred choice wherever sieving time permits.

Two passes are required: one to collect the calibration grid for fitting the coefficients, one to compute C_{EMP} using them.

9.3 Mode C: sampling via primecount

Above the sieving ceiling, $p(n)$ is obtained from primecount at 10^5 points distributed across $[n_0, N]$. No sieve, no storage.

Sampling loses the guarantee: a sample can miss the extremum. Measured across five scales from 2×10^6 to 2×10^7, the ratio of the sampled C_{EMP} to the full value was

Statistic	Value
Minimum	0.9532
Mean	0.9779
Standard deviation	0.0131

Covering the worst observed shortfall requires $1/0.9532 = 1.049$; covering mean $- 3\sigma$ requires 1.065. We use 1.10, wider than both, costing 10% in interval width.

Remark 10 (A failed attempt to make this self-calibrating). We tried computing the shortfall from the current run, using the saturation ratio $C_{\mathrm{EMP}}(50\mathrm{K})/C_{\mathrm{EMP}}(100\mathrm{K})$ as an indicator. Its correlation with the actual shortfall was $r = -0.32$ — weak and of the wrong sign. The margin is therefore estimated from prior measurements, not derived from the run at hand.

9.4 Mode F: extrapolation

Above the sampling ceiling, C_{EMP} is not measured at all. It is extrapolated from a stored series of prior calibrations using a local exponent from the last two scales,

$$C_{\mathrm{EMP}}(N) \approx C_{\mathrm{EMP}}(N_{-1}) \left(\frac{N}{N_{-1}}\right)^{k}, \qquad k = \frac{\ln\left(C_{\mathrm{EMP}}(N_{-1})/C_{\mathrm{EMP}}(N_{-2})\right)}{\ln\left(N_{-1}/N_{-2}\right)}, \tag{23}$$

multiplied by a safety margin (§12).

The local exponent was chosen over a global fit by measurement: mean prediction error 16.1% against 22.9%.

Remark 11 (Mode C was dead code). In an earlier version mode C required the sieved array for its coefficient grid. But mode C's range begins above the sieving ceiling, so whenever it would have been selected the array was absent and control fell through to mode F. Mode C was never chosen. It now builds its grid from primecount exactly as mode F does; the only difference between them is that C measures C_{EMP} by sampling while F predicts it.

9.5 Grid size is computed, not fixed

Modes C and F build their coefficient grid by calling nth_prime, whose cost grows steeply:

n	Cost per call
10^9	5.8 ms
10^{12}	151 ms
10^{13}	622 ms
2^{50}	$\approx$ 22 s

A fixed grid of 20,000 points, chosen without reference to this cost, implies 55 hours at 2^{50}. The grid size is therefore computed at run time from a measured cost of one call and a stated time budget.

The floor of the grid size is justified by measurement. The coefficients are mean-type statistics and saturate quickly:

Grid points	Out-of-training estimation error
50	1,365
100	1,036
200	864
500	880
2,000	974
200,000	1,039

From 100 points upward all values lie in a single band and the variation is not directional. Only at 50 does the estimate degrade clearly. Note that this concerns the coefficients only; C_{EMP}, being a maximum, does require many points, which is why modes C and F handle it separately.

10 Results

10.1 Containment across scales

Every target is at ratio exactly 2.00. True values were obtained from primecount, and independently from a segmented sieve at $n = 2 \times 10^9$ as a cross-check on the tool itself.

N	Target $2N$	Mode	Contained	Edge margin	vs. Axler
5×10^7	10^8	A	yes	—	—
5×10^8	10^9	A	yes	—	—
10^9	2×10^9	A	yes	—	—
5×10^{11}	10^{12}	C	yes	41.7%	116×
2^{40}	2^{41}	F	yes	36.4%	107×
2^{50}	2^{51}	F	yes	47.5%	2,387×
2×10^{15}	4×10^{15}	F	yes	34.7%	2,966×
3×10^{15}	6×10^{15}	F	yes	34.9%	3,608×

Definition 2. The edge margin is $\min\left(p(n) - x_1,\ x_2 - p(n)\right)/(x_2 - x_1)$. A value near 50% means the true value lies near the centre of the interval; a value near 0 means it lies near an endpoint, and the containment was close to failing.

Two features of the table deserve comment.

The ratio to Axler grows with scale. From 116× at 10^{12} to 3,608× at 6×10^{15}. This is not an improvement in the model but a property of the comparison: by (1) the Axler width grows faster than the calibrated interval does.

The edge margins are comfortable and stable. They range from 34.7% to 47.5% with no downward trend as scale increases. Had the calibration been degrading with scale, one would expect these to shrink toward zero. They do not. This is evidence, though not proof, that the extrapolation in mode F is not breaking down over the range tested.

10.2 Comparison with exponentially tight effective bounds

Visser's bounds (2) [9] share a structural feature with the present construction: both take an inverse counting function as the base, and both localise $p(n)$ by bracketing its argument rather than by expanding a series. The difference lies in what supplies the width.

Visser's width comes from a proven effective bound on $|\pi(x) - \mathrm{li}(x)|$ of the form $a\,x(\ln x)^b e^{-c\sqrt{\ln x}}$. Ours comes from a measured maximum deviation. The consequence is the expected one: his interval is unconditional and ours is not.

Remark 12 (An omission we should state). We have not made a numerical comparison. Doing so properly requires fixing the constants (a, b, c) from a specific effective estimate and evaluating both intervals at the same targets; we did not do this. We record it as an omission rather than present a comparison we have not performed.

What can be said without computation is that the two answer different questions. His bound holds; ours is calibrated to observed behaviour. A reader who needs a guarantee should use (2).

A natural question, which we leave open, is whether the empirical calibration described here could be applied with (2) as the base in place of R^{-1}, and whether that would narrow the result further.

10.3 Exhaustive verification at small scales

Sampling a few hundred targets can hide rare failures (§10.4). At scales where it is feasible we therefore tested every integer in the range:

Range	Targets	Containment	Mean width
1.28–2.56×10^6	1,280,000	100.0000%	5,588.9
2.56–5.12×10^6	2,560,000	100.0000%	8,105.2

3,840,000 consecutive targets, no failure, and no case where the intersection search failed to find a valid centre.

Remark 13 (These runs predate three fixes). They were performed with an implementation containing the Axler constant error, the C_{EMP} cutoff error, and the corrupted zeta table (§14). The intervals reported there are therefore inflated by a known factor. The containment rate is unaffected: a wider interval still contains.

10.4 Sampled versus exhaustive targets

The distinction matters more than it might appear. In one configuration during development, an exhaustive scan of 18,005 targets revealed five containment failures. Random samples of 400 targets — the size used for most comparative measurements — had not detected any of them.

A failure rate of $5/18{,}005 \approx 2.8 \times 10^{-4}$ gives a probability of about 11% that a sample of 400 contains at least one failure. Absence from such a sample is therefore weak evidence.

All containment rates in this paper are from exhaustive scans, or from the complete set of targets at that scale.

10.5 Direct jump versus full walk

The model can either walk from N to the target step by step, or jump directly, settling the branch state over a fixed number of steps. The walk is the more conservative procedure; the jump is what makes large scales feasible.

	Targets	Contained	Mean width
Full walk	6,000	6000/6000	947.2
Direct jump	15	15/15	916.8

Ratio 0.968: the jump is 3.2% narrower, with equivalent containment. We had expected the walk to be at least as tight, and it is not; we do not have an explanation for the direction of the difference.

Remark 14. This comparison is at $N = 25{,}000$, small enough that the walk is feasible. Whether the relation holds at larger scales is untested — the walk at $N = 1.28 \times 10^6$ took about four hours.

11 Negative results

This section records approaches that were tried and failed. We include them because several are natural first attempts that look promising until measured, and because two of them (§11.1 and §11.5) bear on the structure of the problem rather than on implementation.

11.1 Alternative normalisations

Since $\ln^2 n$ has no theoretical basis for this use (§5.2), we tested alternatives. The natural criterion is uniformity: a good normalisation makes $|r(n)|/\mathrm{norm}$ roughly constant across the range, so that C_{EMP} is not dominated by one region.

Measured as the ratio of maximum to minimum across four bins of the training range (ideal 1.0):

Normalisation	Dispersion
$\ln^2 n$	1.57×
$n^{0.4}$	1.11×
$\sqrt{n}$	1.10×
$\sqrt{n}/\ln n$	1.08×

$\sqrt{n}/\ln n$ is the most uniform, and it is also the form the Riemann hypothesis suggests for the scale of $\pi(x)$ fluctuations — so it has better motivation than $\ln^2 n$ on both counts.

It performs worse. Measured on actual targets, it widened the interval by a factor of 1.8 (13,914 against 7,846).

Observation 5. Lower dispersion inside the training range does not imply better behaviour outside it. The alternatives are more uniform on $[1, N]$ precisely because they grow faster, and that same growth inflates them on $(N, 2N]$.

Observation 5 is, we think, the most transferable finding in this section. The obvious criterion for choosing a normalisation — fit quality on known data — selects the wrong answer for an extrapolation problem.

Remark 15 (Results do not transfer between architectures). On a modified variant of the procedure, with the branching removed and the fixed envelope substituted for the intersection, $n^{0.4}$ appeared superior. We initially adopted it on that basis and had to reverse the decision. A measurement made on a simplified version of a system does not necessarily hold for the system.

11.2 Percentile extraction

By Observation 3 the interval is set by the two extreme candidates, so improvements to cloud density are invisible. An obvious response is to extract the bounds from a percentile rather than the extremes.

Measured over 18 targets at $N = 2 \times 10^6$:

Extraction	Width	Contained	Min. margin
min/max	7,365	18/18	21.7%
0.5% percentile	7,032	18/18	22.0%
2% percentile	6,578	18/18	18.8%
5% percentile	6,003	18/18	16.0%

The narrowing is real. We did not adopt it, for a reason that is principled rather than empirical: the trimmed candidates were generated by the model as admissible. Discarding them converts the procedure from "the interval covering everything the model admits" to "the interval covering most of what the model admits", which is a different and weaker statement. The measured containment does not distinguish the two at 18 targets.

The option remains in the implementation, disabled by default.

11.3 Self-calibrating the sampling margin

Described in §9: the saturation ratio $C_{\mathrm{EMP}}(50\mathrm{K})/C_{\mathrm{EMP}}(100\mathrm{K})$ was tested as an indicator of how much the sample was missing. Correlation with the actual shortfall: $r = -0.32$, weak and of the wrong sign.

11.4 Probability-weighted branching

A natural extension of the anomaly mechanism is to weight the added candidates by an estimated anomaly probability p and amplitude A, adding candidates at $\pm pA$ rather than at fixed offsets.

	Width	Median error	Surviving candidates
No branching	6,461	677	427
$p = 0.3$	6,524	700	1,268
$p = 0.6$	6,529	741	1,233

The cloud responds strongly — surviving candidates roughly triple — but neither the width nor the median error improves. The branches distribute symmetrically in both directions, so the median does not move and the extremes move outward slightly.

What the approach requires, and lacks, is a direction: a reason to add candidates on one side rather than both. That reduces to predicting where the anomaly lies, which is the subject of the next subsection.

11.5 Predicting anomaly locations

Define an anomaly as a point in the top 1% of $|r(n)|$. Two facts about anomalies are established by measurement (§13): they cluster strongly, and residual persistence is higher inside anomalous regions.

Neither fact yields a predictor of where the next anomaly falls in the unknown region. Six candidate signals were tested against the signed error:

Signal	Correlation with error
$\lvert w\rvert$, the accumulated ridge correction	-0.10
Zeta zeros 61–260 as additional features	-0.036
Disagreement between the $p(n)$ and π models	$+0.001$
Number of candidates surviving the π filter	-0.013
Residual at a single anchor point	decays beyond 10^5
Average over a large window	≈ 0

The first entry is worth its own comment: $|w|$ correlates negatively. The mechanism fires when the model is performing better than usual, which is the opposite of the intended behaviour.

Observation 6. We suggest the obstruction is structural. The anomaly is, by construction, the component the model does not capture. Any signal computed from the model is a function of what the model does capture. There is no reason to expect the latter to inform the former, and measurement finds that it does not.

11.6 A measurement that was an artefact

An early attempt to test whether a branch's state predicts the direction of its error reported $r = +0.70$ for the offset o — an encouraging result that turned out to be arithmetic rather than informational.

Writing the error of branch j as

$$e_j = (\text{base} + o_j + s_j d) - p(n), \tag{24}$$

and noting that base and $p(n)$ are identical across branches at a given n, we have $e_j = o_j + \text{const}$. The correlation of o with e is therefore 1 by construction, and the measured 0.70 merely reflects the contribution of the $s_j d$ term.

The correct test is whether the branch nearest the truth occupies a predictable position in the cloud ordering. Measured across six scales, its normalised rank was

$$0.737,\ 0.316,\ 0.564,\ 0.608,\ 0.555,\ 0.515,$$

mean 0.549, standard deviation 0.125 — consistent with uniform. The nearest branch is in no predictable position.

Remark 16. We record this because the erroneous version was checked and believed before the arithmetic was examined. A correlation of 0.70 between two quantities one of which is defined in terms of the other is not evidence of anything.

11.7 Asymmetric envelopes

If the residual distribution were skewed, an asymmetric envelope would be narrower at equal coverage. It is not:

Percentile	Upper tail	Lower tail	Ratio
99.0	5.149	−5.091	0.989
99.9	6.758	−6.851	1.014
99.99	8.547	−7.740	0.906

Symmetric to within measurement error at every level tested.

11.8 Global envelope widening

If the entire unknown region were treated as anomalous, the envelope would reach its maximum width, which might be argued for as a safety measure. Measured:

Quantity	Change
Interval width	+25.6%
Edge margin	+4.6%

A 5.6-to-1 loss. The arithmetic is immediate: the edge margin is $\min(p(n) - x_1, x_2 - p(n))/(x_2 - x_1)$, and symmetric widening adds a fixed amount to the numerator while adding twice that to the denominator.

11.9 A term that contradicted its own purpose

The base estimator originally included a correction

$$c(n) = a\sqrt{n \ln n} + b, \tag{25}$$

fitted on the training range, intended to centre the model on the known sample.

After the base was upgraded from li^{-1} to R^{-1}, we tested whether it still earned its place. Measured on 7,997 out-of-training targets, with a full ridge refit in each configuration:

Region	With $c(n)$	Without
Inside training (1–5×10^6)	504	**498**
First 5×10^5 beyond	1,200	**930**
6–8×10^6	911	**873**
9–11.5×10^6	1,150	**1063**

Better without it in every region — including inside the training range, which is where it was supposed to help. Overall improvement 8.15%, $p = 1 \times 10^{-47}$ (Wilcoxon signed-rank).

The explanation is specific to the upgrade. With li^{-1} the base carried a systematic bias of order 47,000, and a curve of the form $a\sqrt{n \ln n} + b$ tracked it. With R^{-1} the bias is of order 6,000 and no longer systematic in that shape, so the same curve fits noise. Worse, it was fitted on a 70.4% mask of the points (outliers excluded), whereas c_0 in (13) is computed over all of them — so the two centring mechanisms were not merely redundant but inconsistent.

$c(n)$ was removed. This is the clearest instance in the project of a component that was correct when written and became harmful when something upstream improved.

12 The safety margin

12.1 Why a margin is needed at all

In modes A and B, C_{EMP} is measured over every point of the training range and no margin arises. In mode C it is measured on a sample, and the margin covers what the sample misses — a quantity we could measure directly (§9).

Mode F is different. There C_{EMP} is not measured at all but extrapolated by (23), and the margin must cover the extrapolation error. That error has no measurable ceiling, and this section documents seven attempts to find one.

12.2 Seven approaches, seven failures

1. A constant chosen from observed failures. The first version used 1.55 for small scales and 1.25 for large, chosen after seeing which values had sufficed. When the first genuinely new scale was measured (5×10^{11}), the extrapolation from the prior series undershot by a factor of 1.75 — so 1.55 would not have covered it. A constant fitted to past failures does not anticipate the next one.

2. A saturation indicator. Using $C_{\mathrm{EMP}}(K)/C_{\mathrm{EMP}}(2K)$ within the current run as a proxy for how close the series is to its limit: $r = -0.32$ against the actual requirement, and of the wrong sign.

3. A function of extrapolation distance. The intuition is that a longer extrapolation needs a wider margin. Measured over 146 pairs drawn from two series, the correlation between log(jump) and the ratio (predicted/actual) was $r = +0.49$ — that is, longer jumps tended to overshoot, the opposite of the intuition. Fitting a margin law to this would have widened the interval in exactly the cases where it was already too wide.

4. A function of the anchor scale. $r = -0.31$, weak, and not monotone: the largest required margins occurred in the middle of the range, not at either end.

5. Self-measurement from prior errors. Compute the margin from the worst undershoot observed so far within the series. This works once a large undershoot has occurred and fails before it: measured coverage 8/9, the single failure being the first large jump, where the learned margin stood at 1.10 because every prior error had been positive.

6. Convergence to a limit. If the ratio (predicted/actual) converged, its limit would give the margin. It does not: measured values range over 0.386 to 1.26 with no narrowing as the series lengthens.

7. A Gaussian quantile. Model log(predicted/actual) as normal and set the margin to $\exp(z\sigma - \mu)$. This produces a number and a nominal confidence level, and both are misleading. The normality assumption is unverified, and there is reason to doubt it: C_{EMP} is a maximum-value statistic, and such statistics follow heavier-tailed families. A stated "99% confidence" describes the formula, not the process.

We report a cross-region validation of this approach because it is the only one that came close. Deriving (μ, σ) from scales 10^4–2×10^7 and testing on an independent series 2.5–6.2×10^7 gave zero failures in nine cases at all three confidence levels tested, with the actually required margin there being 1.179 against a derived 1.385. That is genuine cross-region validation — but only for extrapolation jumps of 2.5× or less, which is the operating range and not the regime where the margin is most needed.

12.3 What was adopted

Given that no derivation survived, we use the empirically safest available quantity: the largest ratio observed in paired measurements, where C_{EMP} was both extrapolated and directly measured at the same scale.

Scale	Measured	Extrapolated	Required margin
2×10^6	8.009	10.309	0.777
4×10^6	9.696	12.600	0.770
8×10^6	14.313	12.001	1.193
1.2×10^7	15.237	15.461	0.986
2×10^7	19.224	18.705	1.028
2^{40}	3681.800	2102.565	**1.751**

The margin is 1.751, with no manual inflation. In four of the six cases the extrapolation already overshot, so no margin was required there.

Remark 17 (An inflation that was removed). An intermediate version used 1.85 — the measured 1.751 plus a 6% increment added on the grounds that the measured value looked close to the edge. That increment had no source and has been removed. The margin is now exactly the worst measured value, and it is computed from the table above at run time, so adding a new paired measurement updates it automatically.

12.4 The weakness we cannot resolve

Five of the six paired measurements lie between 2×10^6 and 2×10^7. The sixth, at 2^{40}, is the worst of them — and it is the only one at a large scale.

Two readings are consistent with the data: that the requirement is roughly constant and 2^{40} was an unlucky draw, or that it grows with scale and 2^{40} is the beginning of a trend. Distinguishing them requires paired measurements at 10^9, 10^{11}, 10^{12}, each of which costs hours. We flag this as the most consequential open item (§15).

13 Ancillary measurements

The following were obtained while investigating the residual structure. They are not required by the construction, and we report them for their independent interest.

13.1 Anomalies cluster

Defining an anomaly as a point in the top 1% of $|r(n)|$:

Quantity	Value
Anomalous points adjacent to another anomaly	93.6%
Expected under random assignment	1%
$P(\text{anomaly at } n+d \mid \text{anomaly at } n)$, $d = 10^3$	0.838
Baseline rate	0.010
Ratio	84×

The clustering is strong and unambiguous. It is also, by §11.5, not usable for prediction: knowing that anomalies cluster does not locate the next cluster.

13.2 Persistence: the Hurst exponent

Rescaled-range analysis of the residual series gives $H = 0.348$.

Since $H < 0.5$, the series is mean-reverting rather than persistent: a large positive residual is more likely than chance to be followed by a negative one. This is consistent with the residual being an oscillation around a correct trend, and inconsistent with any drift.

Remark 18. This result also closed a design question. An earlier version of the anomaly mechanism included a "rebound" phase, on the assumption that a large residual would be followed by one of opposite sign at a characteristic distance. Direct measurement over 10^7 points found that residuals persist at short range (§13.3) and revert only over long ones, with no characteristic rebound distance. The phase was based on behaviour not present in the data.

13.3 Short-range autocorrelation

Distance d	$\rho(d)$
10^3	0.995
10^4	0.964
10^5	0.721
10^6	−0.171
5×10^6	+0.001

Strong at short range, gone by 10^6. The negative value at 10^6 is within estimation noise (standard error about 0.058 at the sample sizes used).

Conditional on anomaly state. Persistence is higher inside anomalous regions:

Distance	Inside anomaly	Outside
10^4	0.954	0.858
3×10^4	0.836	0.673
6×10^4	0.41	0.00

13.4 The online correction

Because $p(n)$ is known exactly up to N, the residual at N is known for free. By §13.3 it is informative about residuals a short distance beyond. The optimal linear predictor is

$$\hat{r}(N+d) = \rho(d)\, r(N), \tag{26}$$

with ρ selected according to whether the anchor lies in an anomalous region.

Measured against the uncorrected estimate, 400 targets per distance:

d	Uncorrected	Corrected	Gain	p (Wilcoxon)
10^4	1,095	1,028	6.1%	0.005
3×10^4	1,959	1,759	10.2%	5×10^{-4}
6×10^4	2,845	2,424	14.8%	4×10^{-13}
10^5	3,350	2,729	18.5%	4×10^{-15}

Beyond 10^5 the gain vanishes, and (26) returns the uncorrected estimate automatically since $\rho \to 0$.

Remark 19 (This is the correct form of an idea that first failed). An earlier attempt added an independent anomaly term pA on top of (26), and degraded the estimate by 57.9% at $d = 2000$. The reason is that (26) is already the optimal linear predictor given $r(N)$; anything added to it double-counts. The correct use of the anomaly information is not to add a term but to select a different ρ, which is what the table above does.

Remark 20 (Guarding against spurious negative ρ). The first implementation interpolated ρ across all measured distances, including a spurious $\rho = -0.331$ at $d = 3 \times 10^5$ that was noise. This produced an 8.6% degradation at $d = 2 \times 10^5$. The current version truncates at the first distance where $|\rho|$ falls below a trust threshold and zeroes everything beyond, so the correction switches off cleanly instead of doing harm.

13.5 A range law

Combining §13.3 with the residual magnitudes suggests that the distance over which an observed residual remains informative scales as

$$d_{\max} \approx \left(\frac{r}{\ln x}\right)^2. \tag{27}$$

This is an empirical fit, not a derivation. It is consistent with the measured decay of ρ across the scales tested.

14 Implementation defects

Several defects produced effects large enough to invalidate conclusions while leaving the containment rate intact — that is, silently. We record them in full, since the pattern is instructive.

Defect	Measured impact
Axler upper constant set to 0 instead of 10.667	Interval 13.7× wider than necessary at $n = 10^7$
C_{EMP} cutoff at $n \geq 10$ (§5.3)	C_{EMP} inflated 8.5×
Corrupted zeta-zero table	One entry (109.18895) is not a zero; one genuine zero (111.029536) absent
Autocorrelation measured after subsampling (§7.5)	Measured -0.036 where the truth was 0.849
π-side decay inconsistent with $p(n)$-side	Traversed 10^7 points to obtain a value that 12,000 points give to 2.7×10^{-5}
Memory term cancelling itself (§7)	Decayed to zero within ten checkpoints; no effect
$c(n)$ retained after the base upgrade (§11.9)	Error +8.15%, $p = 1 \times 10^{-47}$

Defect	Measured impact
Convex hull allocating $O(N)$ float arrays	Out-of-memory at $N = 5 \times 10^8$
Gap $2 \to 3$ is odd	Corrupted the compressed prime representation
Misaligned calibration series	C_{EMP} for the π side 1,278× too small

14.1 The silent failure

The last entry merits detail, since it is the clearest example of the pattern.

Two parallel series — calibration values for the $p(n)$ side and for the π side — were filtered by different conditions and then truncated to a common length. The truncation dropped the last element of one series, so every value in it was paired with a scale 24,000 times smaller than its own. The extrapolated exponent collapsed, and C_{EMP} for the π side came out 1,278× too small.

The consequence: the π cloud narrowed to near-nothing, the intersection (16) became vacuous, and the reported interval was determined entirely by the Axler clipping (17).

Containment still succeeded — the Axler bound being unconditional — so the run appeared healthy. The only symptom was that the reported ratio to Axler had fallen to 1.0, meaning the calibrated component was contributing nothing.

Remark 21. Three runs totalling about nine hours were performed before this was noticed. The lesson is specific: when a system has a proven fallback, the fallback masks failure in the component under test. A ratio of 1.0 against the fallback should be treated as an error condition, not as a result.

15 Open questions

15.1 Does the required margin grow with scale?

The most consequential unresolved question (§12). Five of six paired measurements are at 2–20×10^6; the single large-scale point, at 2^{40}, is also the worst. Resolving this requires paired measurements at 10^9, 10^{11} and 10^{12} — each a run of hours, but entirely feasible.

If the requirement grows, mode F at large scales is under-protected and the margin should be a function of scale rather than a constant. If it does not, the current constant is adequate and the 2^{40} point was a draw from the tail.

15.2 Is the anomaly location predictable at all?

Six approaches failed (§11.5), and we offered a structural reason (Observation 6). But that reason, if correct, implies something stronger than we have shown: that no quantity derived from the model can predict its own failures. That is a claim about the class of estimators, and we have tested six members of it.

An estimator built on information the current model does not use — the higher zeros in a different parameterisation, or local structure in the prime gaps themselves — is not excluded by our measurements.

15.3 Why is the jump narrower than the walk?

The direct jump gives intervals 3.2% narrower than the full walk, with equivalent containment. We expected the reverse. The comparison was made at a single small scale where the walk is feasible, and we have no explanation for the direction of the difference.

15.4 Would $\pi(x)$-based grids raise the ceiling?

All modes are bounded by the nth_prime limit of the available tool, 2.16×10^{17}, because the calibration grid is built by inverse queries. The forward function $\pi(x)$ reaches considerably further. Rebuilding the grid in terms of $\pi(x)$ and inverting locally would raise the ceiling, at the cost of an extra inversion per grid point. We have not implemented or costed this.

15.5 Is $\ln^2 n$ the right normalisation?

Section 11.1 shows it outperforms the alternatives tested, and §5.2 shows it has no theoretical claim to the role. Both can be true. A normalisation derived from the residual structure rather than borrowed from the gap literature might do better; we did not find one.

16 Conclusion

The procedure yields an empirically calibrated interval for $p(n)$, 107–3,608 times narrower than the tightest proven bounds, with a measured containment rate of 100% over eight scales up to 6×10^{15} and over 3.84 million exhaustively tested targets at smaller scales.

It is not a theorem, not a faster algorithm for computing $p(n)$, and not a guarantee. Its contribution is a direct measurement of how far analytic estimates deviate in practice, an interval calibrated to that measurement, and a documented account of which approaches to tightening it do and do not work.

The defensible form of the claim is: an empirical interval, narrower by a measured factor than proven bounds, with a containment rate measured over a stated number of targets, and no theoretical guarantee.

A Parameters and their basis

Parameter	Value	Basis
Zeta zeros K	60	Measured; more degrades (§4.5)
C_{EMP} cutoff n_0	10,000	Measured; stable from here (§5.3)
Normalisation	$\ln^2 n$	Measured against alternatives (§11.1); no theoretical basis (§5.2)
Envelope multiplier	± 2	Design choice, not optimised
Margin floor	1.751	Computed from paired measurements (§12)
Sampling margin (mode C)	1.10	Covers worst measured shortfall plus safety (§9)
Checkpoint spacing CP_{pn}, CP_{π}	40, 5	Design choice
Branch cap / retained ($p(n)$)	128 / 64	Design choice
Branch cap / retained (π)	1024 / 512	Design choice
Settling steps	800	Design choice
Boundary seeds	5	Sufficient for stability (§6.4)
Grid time budget	900 s	User-set; grid size derived (§9.5)
Axler upper constant	10.667	[1], Thm 1.1
Axler lower constant	11.508	[1], Thm 1.2

Parameters marked "design choice" were set at the outset and not subsequently optimised. We do not claim they are optimal; we claim only that the results reported were obtained with these values.

B Code and data availability

The implementation, eight experiment scripts with embedded true values for automatic verification, and an independent benchmarking tool are available at [repositoryURL].

Each experiment script is self-contained and reproduces one row of the table in §10. The benchmarking tool measures the limits and per-call costs of primecount on the reader's own hardware, and verifies every true value used in this paper.

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